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Design smarter, not harder: Evolving design reuse methodologies in EDA

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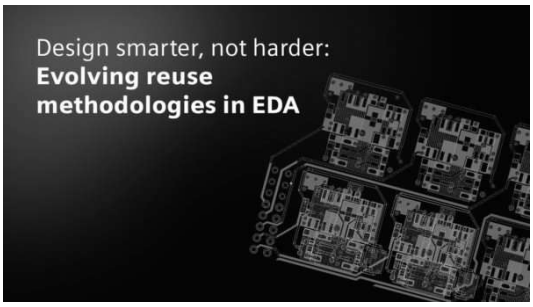

By Arlina Yang

In this episode of the *Printed Circuit Podcast*, host Steph Chavez explores the growing impact of **design reuse** in electronic design automation (EDA), joined by two longtime collaborators and Siemens EDA experts—**Andre Mosley**, Marketing Development Specialist, and **Carlos Gazca**, Senior Technical Marketing Engineer.

Together, they connect the dots on how design reuse has evolved from scattered, informal copy-paste strategies to structured, data-managed ecosystems — reducing time-to-market, increasing quality, and preserving valuable engineering IP.

From manual chaos to managed confidence with design reuse

Early in their careers, Chavez and Gazca found themselves working late-night design shifts, manually handing off files and relying on zip folders and shared drives. "It wasn't elegant," Gazca recalled. "There were versioning issues, sync issues — it was copy and paste with fingers crossed."

Design reuse existed, but it was chaotic. Designs were hard to trace, edits hard to control, and engineers couldn't confidently track where a circuit came from or what version was used.
"That's not reuse. That's duplication," said Chavez. "True reuse is version-controlled, traceable, and integrated into your design data ecosystem."

The driving force: Customers demanding efficiency

Ironically, it wasn't always engineering teams that initiated design reuse. As Chavez explained, the push often came from customers. This customer-driven pressure to reduce costs and shorten product development cycles pushed organizations to formalize reuse strategies. And with the right tools, the payoff became undeniable.

"With data management," Gazca noted, "I can search parametric data — say I need a 5V buck converter—and reuse it instantly."

Engineers can now access simulation results, requirements, and constraints tied to reusable blocks—giving confidence to tweak or version based on real, known-good circuits.

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"With a modern platform, you can track which blocks are reused across how many designs, by whom, and how often," said Mosley. "Some companies even incentivize engineers based on reuse metrics."

"You're not guessing if the circuit will work in the lab," Chavez explained. "It's been proven. You're designing quality in — you're not checking for it later."

Design reuse isn't plug-and-play

Nonetheless, formal design reuse does not come easily. To decide how many reusable blocks to make, teams have to invest in an initial strategy to decide how big to make them, whether it is a complete subsystem or a single power stage.

"We've gone from passing circuits by email to controlling them through secure, collaborative platforms," said Mosley. "And that's essential for global teams and sensitive IP."

Buy-in related to culture is also critical, since engineers might not promote the usage of designs that were not created by them, and trust and visibility are crucial in this situation. Lastly, the whole process carries a robust data infrastructure, which allows teams to monitor the usage, preserve compliance, and protect reused design integrity.

Training the next generation of designers

The conversation also touched on how young engineers can grow with reused designs.

"Reuse isn't just a tool feature — it's a design mindset," said Chavez. "You need to learn board fundamentals, master your toolset, and embrace reuse as part of design quality —not a shortcut."

Gazca shared how, early in his career, he learned by browsing old circuits in the company library. "It was like Googling an idea: I'd see what others had done and build from there."

In the future, reuse, AI, and automation will transform the EDA into an even smarter tool. A security-first model will be needed, which allows teams to follow who, where and what was visited, which is essential in an international, controlled world.

"In the future, the most competitive companies will be the ones who treat their **design data** as their most valuable IP," Mosley emphasized. "Reuse isn't optional anymore — it's strategic."

The reused blocks will also assist engineers to estimate cost and optimization of design real estate more effectively. In addition, the integration of AI will bring design reuse to a new level, and in the future, systems will be suggested to propose reusable blocks automatically depending on the system-level objectives and simplify the design process, as well as decision-making.

Conclusion: Design Reuse Is the Future of Efficient Design

From zip files and copy-paste chaos to a structured, intelligent design ecosystem—reuse in EDA has come a long way. "It's not about cutting corners," said Chavez. "It's about designing smarter — faster — with less risk and more confidence."

As Gazca put it: "When reuse is done right, the first time you power up your board — it works."

Learn more expert advice on design reuse, and listen to the Printed Circuit Podcast.

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Arlina Yang

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